SOFTWARE ENGINEERING · MINI PROJECT

ReturnLoop

Software requirements & design document

SYSTEM	Returns and reverse-logistics centre for an e-commerce business
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INSTITUTE	SVKM's NMIMS, Hyderabad
VERSION	1.0 · October 2026
LIVE APP	return-loop-c25e.vercel.app



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01 Introduction

1.1 Purpose

ReturnLoop gives an online store one place to run every return: from the customer's request and the doorstep pickup, through quality check and grading, to a correct refund and the item going back on the shelf, to the repair bench or to scrap. Every step is recorded.

1.2 Scope

IN SCOPE

- Six roles and a 13-status return workflow
- Eligibility rules, pickup booking with two attempts
- QC checklist, A / B / C grading, fraud check
- Refurbishing and re-inspection
- Refund calculation and inventory log
- Admin dashboard and frequent-returner flags

OUT OF SCOPE

- Real payments or a payment gateway
- Passwords and real accounts (a role switcher stands in)
- Courier API integration

1.3 Users and roles

ROLE	DEMO USER	MAIN TASKS
Customer	Chirag	Requests returns, books pickups, tracks and closes them
Pickup agent	Ayush Dubey	Collects parcels, or marks the customer absent
Warehouse	Ansuman Singh	Checks parcels in at the returns centre
QC inspector	Aadi Srivastava	Inspects, grades A / B / C, rejects mismatched items
Refurbisher	Anumeet	Repairs grade B items for a second check
Admin	Aryan Singh	Watches totals, reasons, grades and flagged customers

1.4 Key terms

Return code
The ID of a return, e.g. `RL-2026-0004`.

Grade A / B / C
QC's verdict: A restocked (100%), B repaired then rechecked (70%), C scrapped (0%).

Handoff
After each step the app names the next desk and opens its page in one click.

Return window
7 days from delivery; outside it, the item can't be returned.

02 Functional requirements

What the system must do. Priority uses MoSCoW: **Must** is needed for a working return flow, **Should** makes it complete.

ID	REQUIREMENT	DESCRIPTION	ACTOR	PRIORITY
FR01	**Role switch**	Choose any of 6 roles and a user without a password; each page rejects the wrong role (403).	All	MUST
FR02	**View orders**	Customer sees delivered items with eligibility, or the reason it can't be returned.	Customer	MUST
FR03	**Eligibility**	Return allowed within 7 days of delivery, for returnable products, with no other open return.	System	MUST
FR04	**Request return**	Reason, comment (≤ 300 chars), optional photo (PNG/JPG/WEBP, ≤ 3 MB).	Customer	MUST
FR05	**Book pickup**	Date from tomorrow to +5 days, slot Morning / Afternoon / Evening; reschedule.	Customer	MUST
FR06	**Collect parcel**	Agent marks picked up or absent; the 2nd absence cancels the return.	Pickup agent	MUST
FR07	**Receive**	Warehouse checks the parcel in; it joins the QC queue.	Warehouse	MUST
FR08	**Inspect & grade**	QC checklist, suggested grade, A/B/C; a mismatched item is rejected as fraud.	QC inspector	MUST
FR09	**Refurbish**	Grade B goes to repair, then re-inspection: pass → restock (70%), fail → scrap.	Refurbisher, QC	MUST
FR10	**Refund**	final = max(0, base × grade% − pickup fee), damaged/wrong-item exception, created once.	System	MUST
FR11	**Track**	Live status (polled every 5 s), history timeline and refund breakdown.	Customer	SHOULD
FR12	**Dashboard**	Totals, reasons and grades charts, frequent-returner flags (> 3 in 30 days).	Admin	SHOULD
FR13	**Hand off**	After each step, a card names the next desk and opens its page in one click.	All	SHOULD

03 Non-functional requirements

How well the system must do it, and how each quality is checked.

ID	QUALITY	REQUIREMENT	HOW IT IS CHECKED
NFR01	**Performance**	60 fps on a mid laptop, 30+ on phones; pixel ratio capped; 3D pauses off-screen.	Frame-rate guard halves particles below 40 fps
NFR02	**Reliability**	Refund + inventory log + status change in one transaction; no double refunds.	Unique refunds.return_id; double-submit tests
NFR03	**Security**	Role guard on every blueprint; customers only see their own returns; upload validation.	403/404 tests; file-signature check on photos
NFR04	**Usability**	Plain-language errors and empty states; the handoff card always names the next step.	End-to-end relay test over HTTP
NFR05	**Accessibility**	WCAG AA contrast, keyboard-only use, visible focus, reduced-motion mode, text on every badge.	Manual keyboard pass; reduced-motion mode
NFR06	**Maintainability**	Logic in services/, rules in config.py, thin routes; a pytest suite over services, pages and the full flow.	pytest on every increment
NFR07	**Portability**	Python 3.10+, SQLite, no build step: install, seed, run in 3 commands.	Runs locally and on Vercel
NFR08	**Robustness**	Static fallbacks without WebGL or internet; friendly 403 / 404 / 413 pages.	Pages tested with 3D unavailable

04 Analysis & design diagrams

Each diagram below is drawn from the code: the statuses and moves come from `services/state_machine.py`, the classes from `models.py`, and the messages from the real functions.

4.1 Use case diagram

The six actors stand outside the system boundary, each linked to the use cases they can start. The customer views orders, requests a return, books a pickup, tracks it and finally closes it; requesting a return always *includes* Check eligibility (the 7-day window, a returnable product, no open return). The pickup agent, warehouse, QC inspector and refurbisher each handle one stage of the parcel's journey. Grading and re-inspecting always include Issue refund, so no graded item is left without one. The admin only reads: the dashboard and the list of flagged customers.

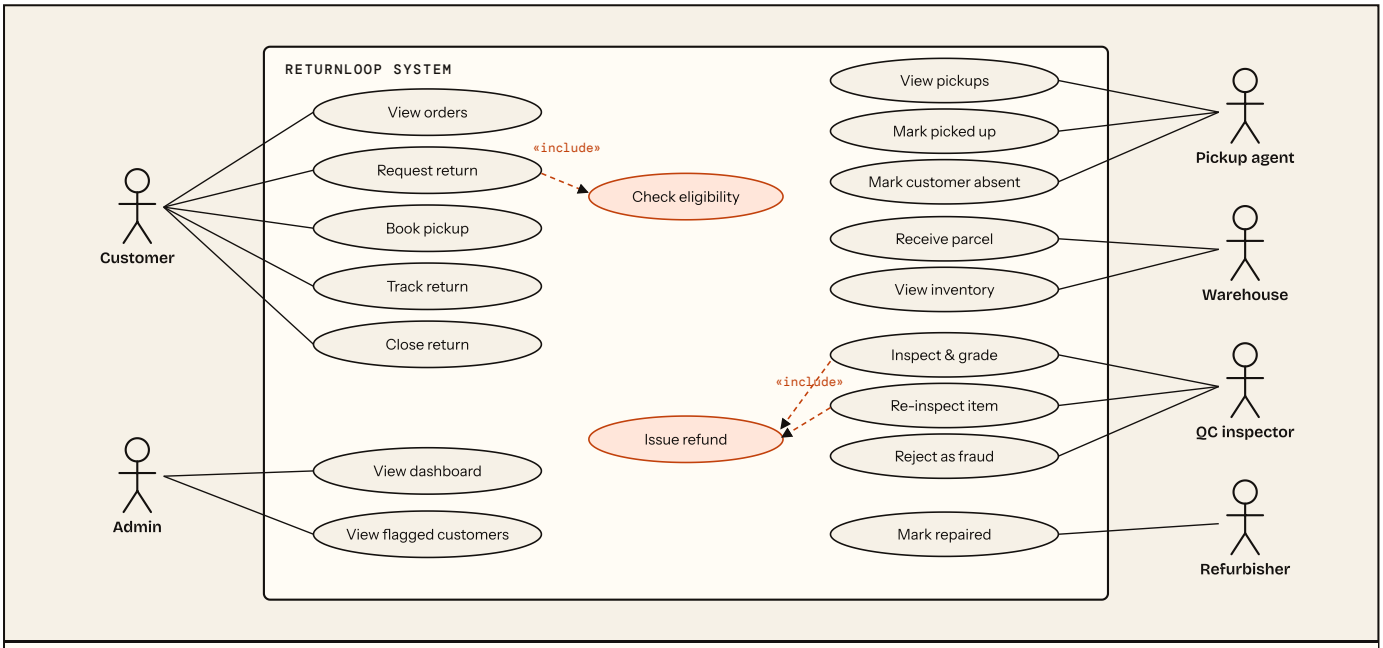


FIGURE 4.1 Use case diagram

4.2 Activity diagram

The diagram follows one return from start to finish. The customer opens their orders; an ineligible item only shows the reason, and the flow ends there. An eligible item is requested and a pickup booked: if nobody is home the agent rebooks for the next day, and a second miss cancels the return. A collected parcel is received at the centre, where QC first checks it matches the order, otherwise it is rejected as fraud. Grade A restocks and C scraps the item; B goes to the refurbisher and a recheck decides between restock and scrap. Every route ends in the same refund and inventory log, and the customer closes the return.

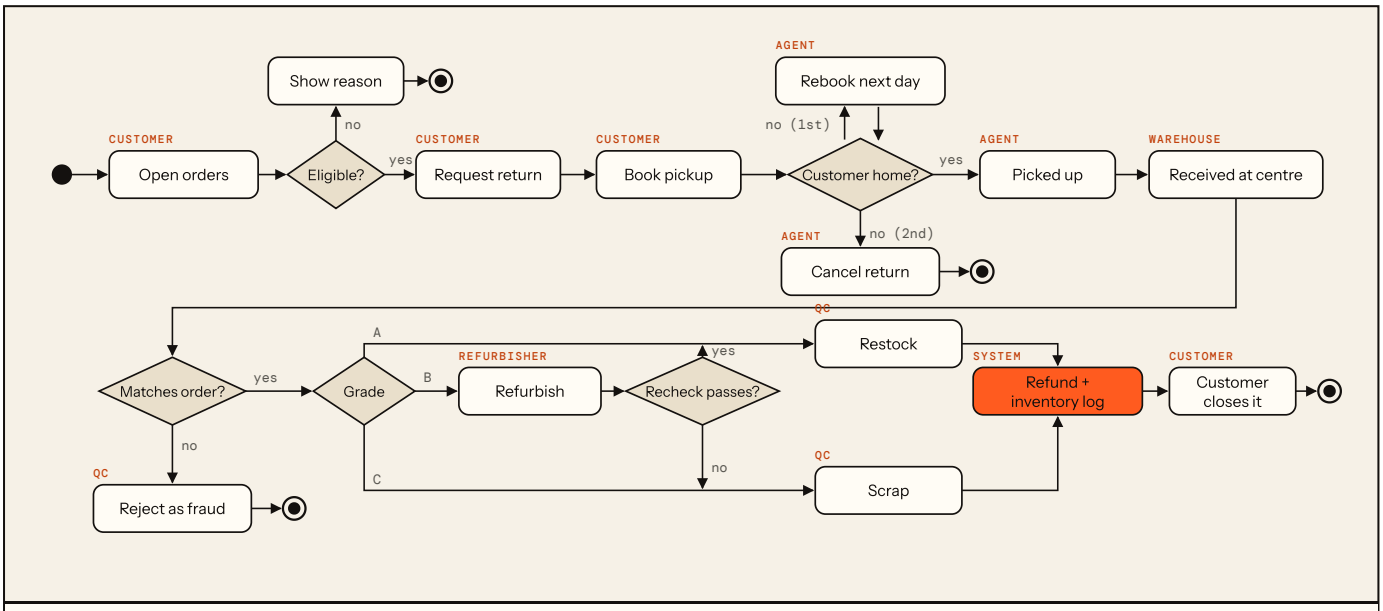


FIGURE 4.2 Activity diagram: one return, start to finish

4.3 Swimlane diagram

This is the same process split into one column (lane) per role, so it shows who is responsible for each action. Reading top to bottom, the return starts in the customer's lane and moves to the pickup agent, the warehouse and then the QC inspector. Every arrow that crosses into another lane is a handoff, which the app turns into a "Continue as ..." card. The refurbisher's lane is used only for grade B items, which loop back to QC for a recheck. The System lane holds the automatic steps (routing by grade, restocking or scrapping, issuing the refund) before the refund arrives back in the customer's lane.

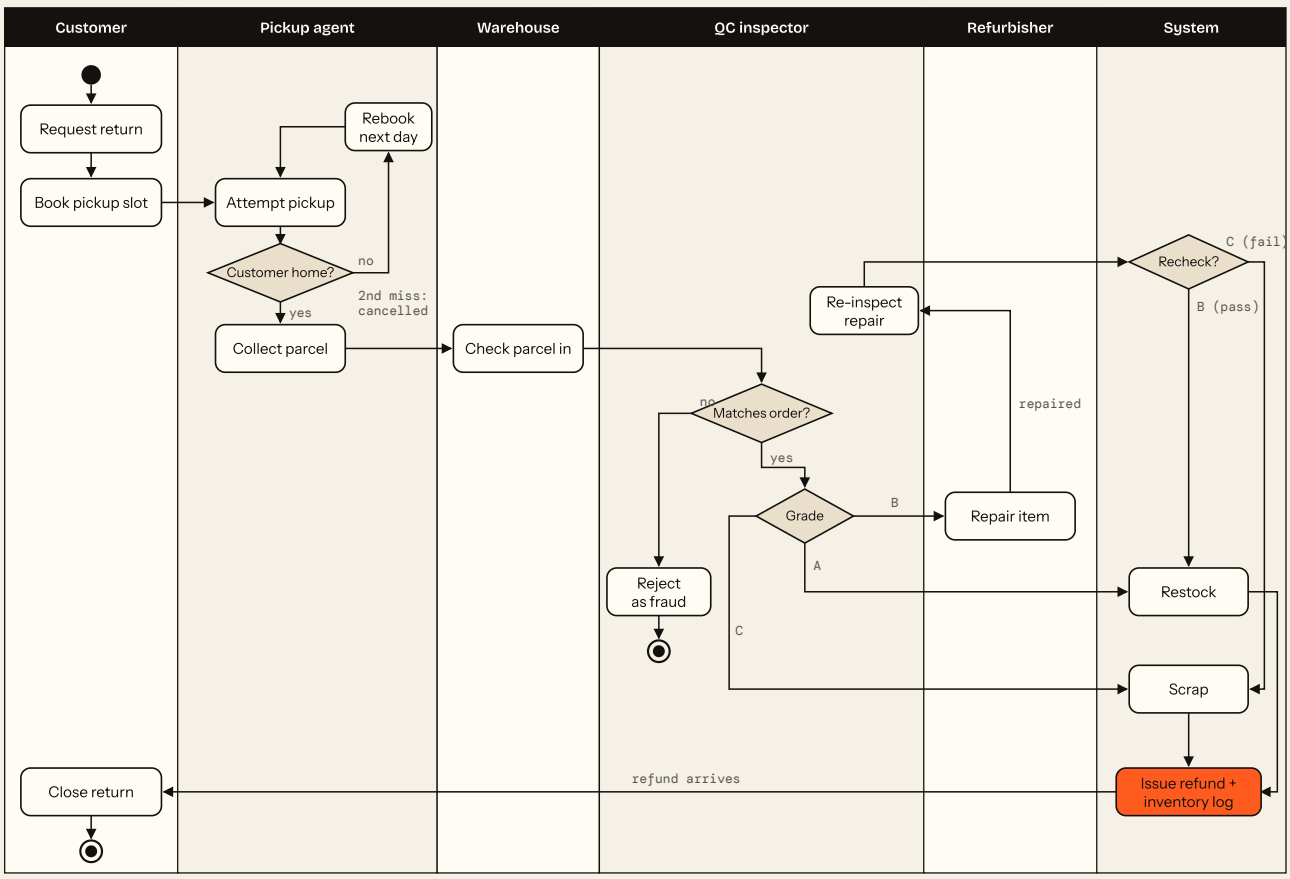


FIGURE 4.3 Swimlane activity diagram

4.4 Class diagram

The domain model has eleven classes, one for each database table in `models.py`. A Customer places Orders, each Order contains one or more OrderItems, and each item refers to a Product. A ReturnRequest belongs to one OrderItem and sits at the centre of the model: it owns its Pickups, QC reports, refurbish jobs, its Refund and its StatusHistory. The filled diamonds mean composition, so those parts cannot exist without their return; InventoryLog rows only reference the return they came from. The multiplicities (1, *, 0..1) match the database rules, for example a return has at most one refund.

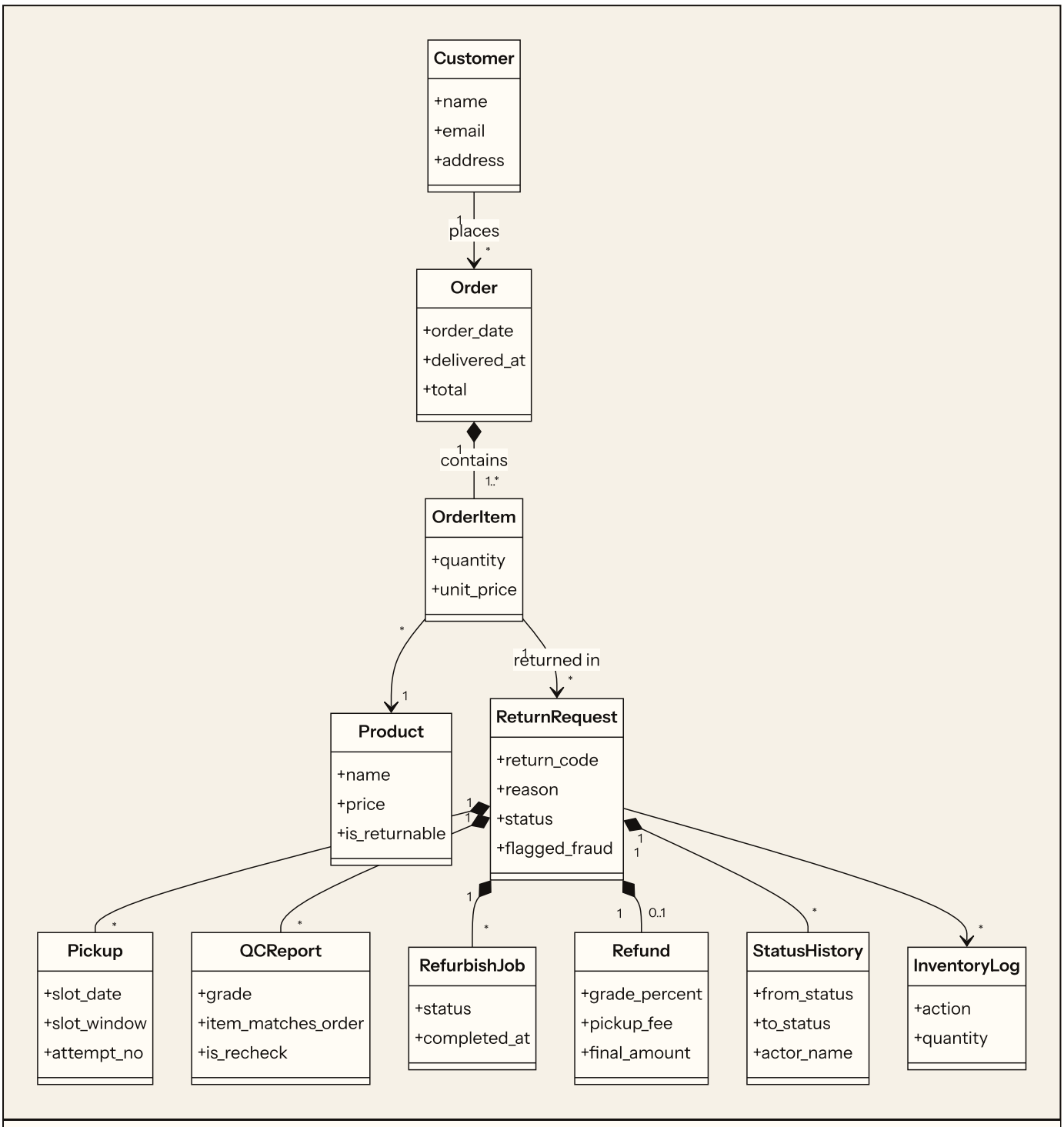


FIGURE 4.4 Class diagram

4.5 CRC cards

Each card names a class or service, its responsibilities (what it knows and does) and its collaborators (the classes it works with). The model classes, such as ReturnRequest, Pickup and QCReport, mainly hold the data for one step of a return. The services hold the rules: the state machine allows only valid moves, Eligibility checks the return window, and the refund service computes the amount exactly once. Handoff and Flags read across several classes to decide who acts next and which customers return too often. Keeping rules in services and data in models is what keeps the route handlers thin.

ReturnRequest

`models.py`

RESPONSIBILITIES

- Holds reason, comment, photo and status
- Knows its pickups, QC reports, refund and history
- Gives its current pickup and latest report

COLLABORATORS

- OrderItem
- Pickup
- QCReport
- Refund
- StatusHistory

State machine

`services/state_machine.py`

RESPONSIBILITIES

- Allows only the moves in the transition table
- Writes a history row for every move
- Refuses anything else

COLLABORATORS

- ReturnRequest
- StatusHistory

Eligibility

`services/eligibility.py`

RESPONSIBILITIES

- Checks the 7-day window
- Checks the product is returnable
- Checks there is no open return

COLLABORATORS

- OrderItem
- Product
- ReturnRequest

Pickup

`models.py`

RESPONSIBILITIES

- Holds slot date and window
- Counts attempts (max 2)
- Records the outcome

COLLABORATORS

- ReturnRequest
- Pickup agent

QCReport

`models.py`

RESPONSIBILITIES

- Holds the checklist and grade
- Marks a recheck
- Keeps the inspector's remarks

COLLABORATORS

- ReturnRequest
- QC inspector

RefurbishJob

`models.py`

RESPONSIBILITIES

- Tracks open / repaired
- Keeps repair notes and technician

COLLABORATORS

- ReturnRequest
- Refurbisher

Refund service	services/refund.py
RESPONSIBILITIES	COLLABORATORS
• Computes base × grade% − fee • Applies the damaged / wrong-item exception • Creates the refund and inventory log once	• ReturnRequest • QCReport • Refund • InventoryLog • State machine

Handoff	services/handoff.py
RESPONSIBILITIES	COLLABORATORS
• Works out who acts next • Picks the page they land on • Lists returns each desk handed on	• ReturnRequest • Pickup • RefurbishJob

Flags	services/flags.py
RESPONSIBILITIES	COLLABORATORS
• Counts returns per customer in 30 days • Flags more than 3	• Customer • ReturnRequest

4.6 Data flow diagrams

Level 0 treats ReturnLoop as a single process (0) and shows only the data flowing between it and the six external entities. The customer sends a request, photo and pickup slot and gets back eligibility, status and the refund breakdown; each staff role exchanges its work list and its results. **Level 1** opens that process into eight numbered processes, from 1.0 Check eligibility to 8.0 Report & flag, and six data stores D1–D6. Data enters from an entity, is changed by a process and written to a store: 2.0 Schedule pickup writes D3 Pickups, which the agent then reads as the pickup list. The dotted line shows that every status change is also written to D6 Status history.

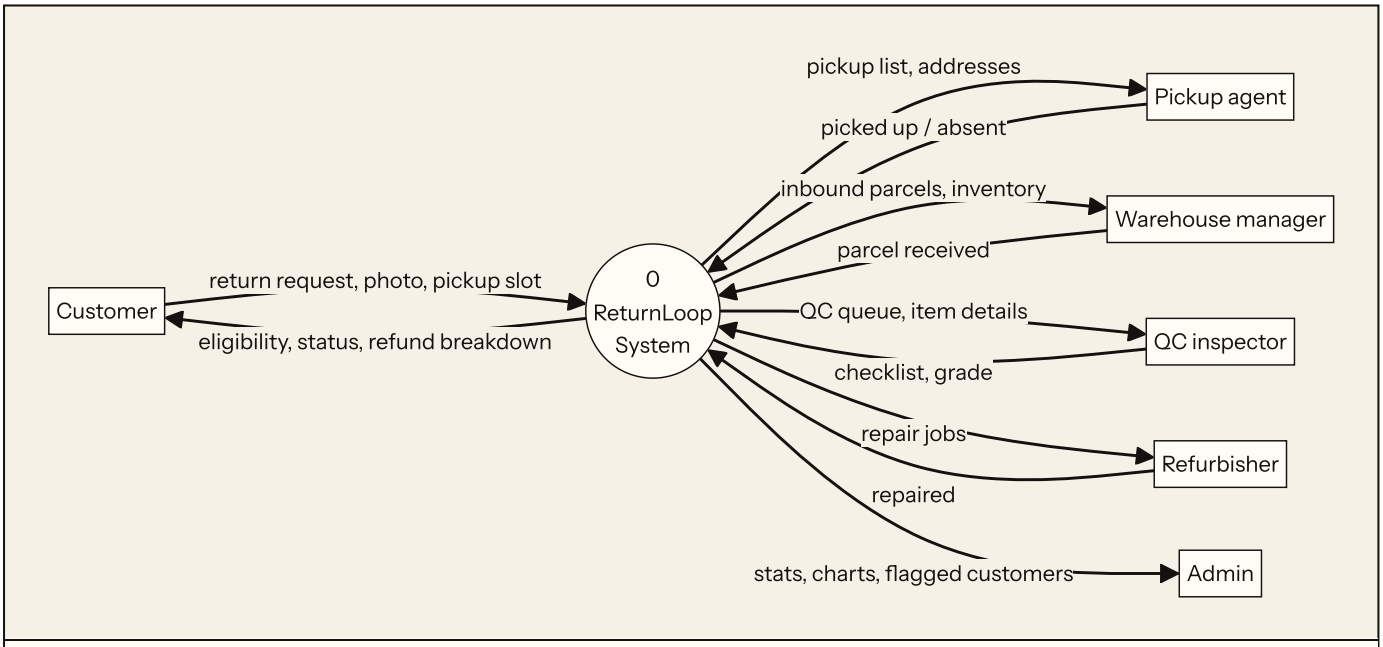


FIGURE 4.6a DFD level 0 (context diagram)

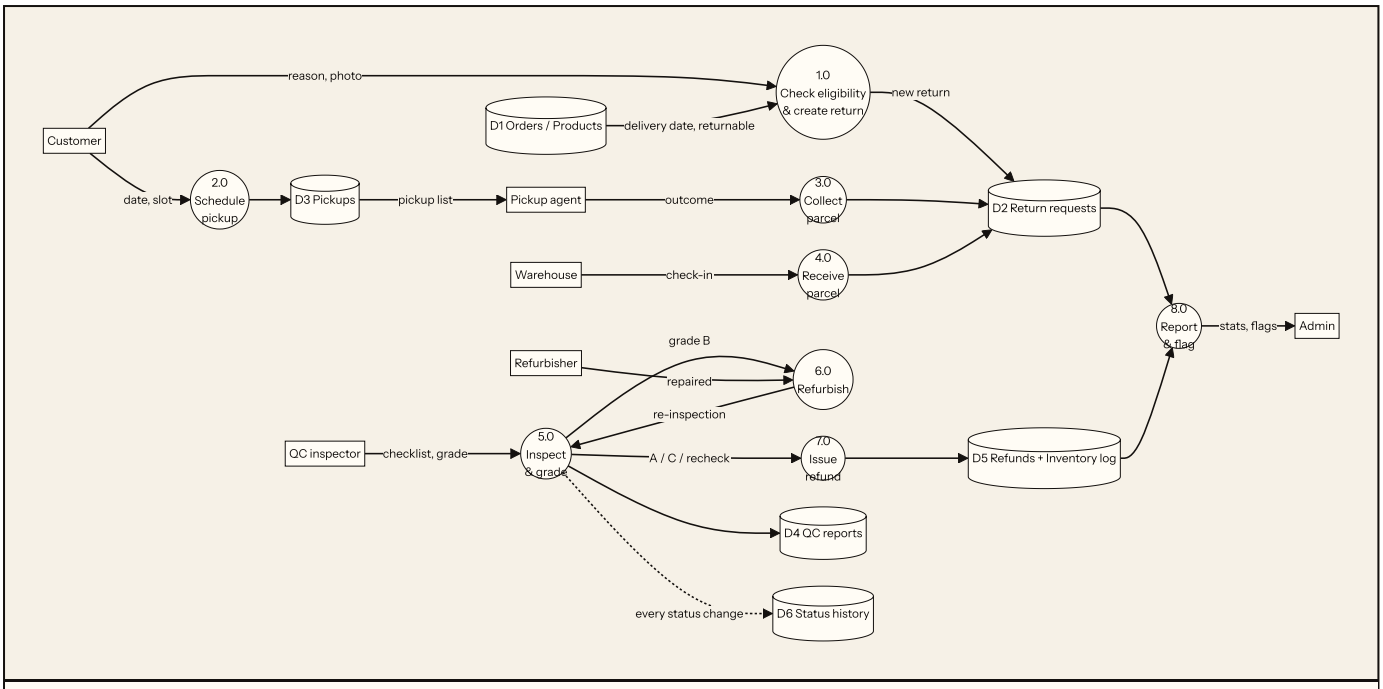


FIGURE 4.6b DFD level 1: processes and data stores

4.7 State diagram

A return is always in exactly one of 13 states. It starts as Requested and moves through Pickup Scheduled, Picked Up and Received; it can stop early as Cancelled (a second missed pickup) or Rejected (not eligible, or the item doesn't match the order). After grading it is Restocked, sent through Refurbishing and Re-inspection, or Scrapped; both Restocked and Scrapped lead to Refunded and then Closed. Each box lists what happens on entering the state (`entry/`) and while in it (`do/`), and each arrow reads `event [guard] / action` , e.g. `inspect [mismatch] / flagFraud` . `change_status()` refuses any move not drawn here, so the diagram is also the rule.

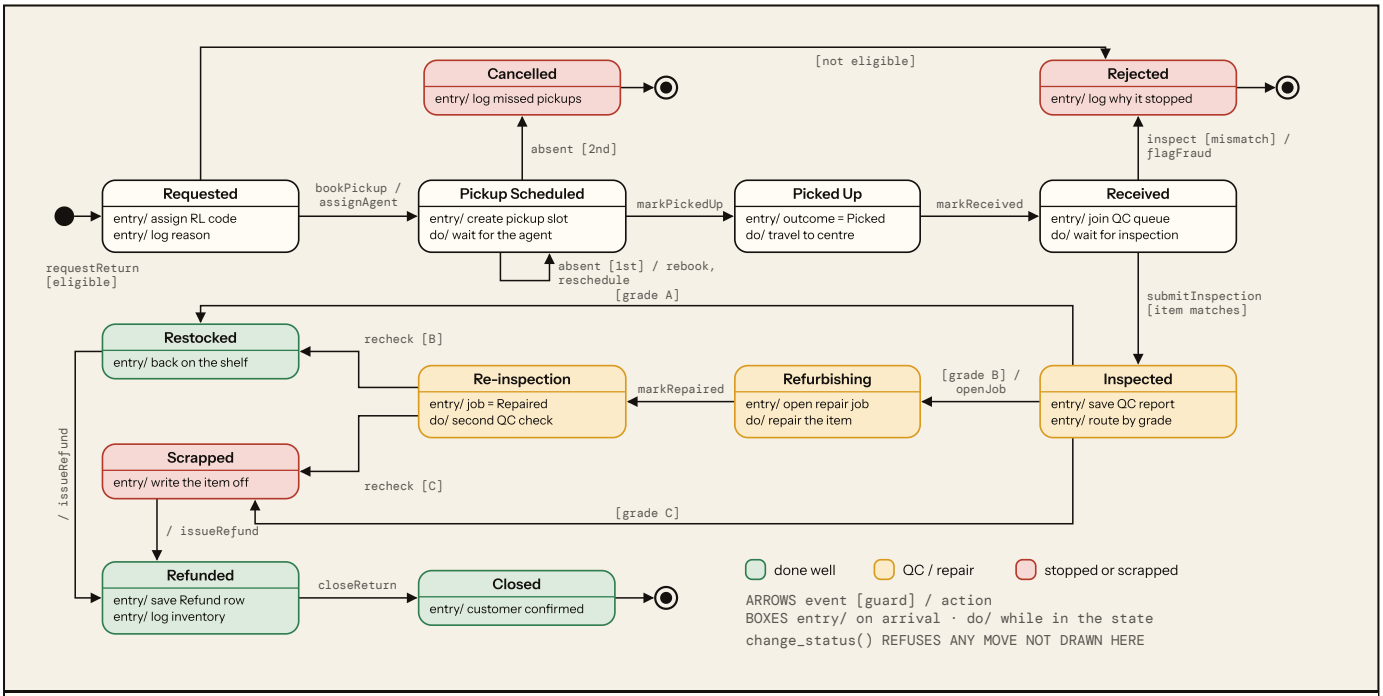


FIGURE 4.7 State diagram of a return

4.8 Sequence diagrams

Both diagrams show objects as columns and messages in time order from top to bottom. **(a)** When the QC inspector grades a return A, the browser posts the form to the qc blueprint, which calls `submit_inspection()`; that moves the return to Inspected and then Restocked through the state machine and calls `issue_refund()`. The refund service calculates the amount, moves the return to Refunded and writes the refund, inventory log and history rows in one commit, so either all of it is saved or none. **(b)** After the agent marks a parcel picked up, `hand_off()` remembers the return; the next page asks `services.handoff` who is next, and "Continue as warehouse" posts to `/switch`, which opens the warehouse page with that parcel highlighted.

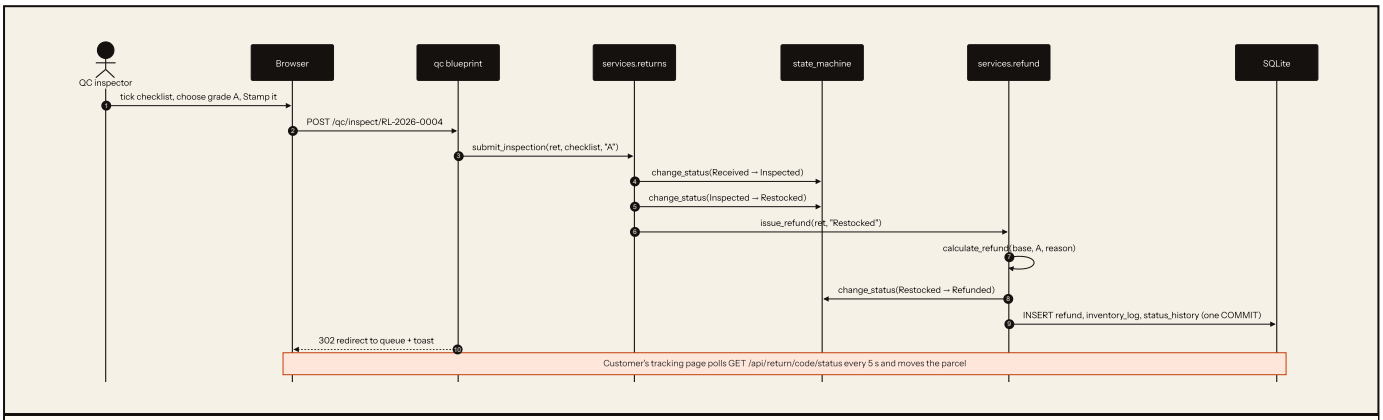


FIGURE 4.8a Sequence: QC grades a return and the refund is issued

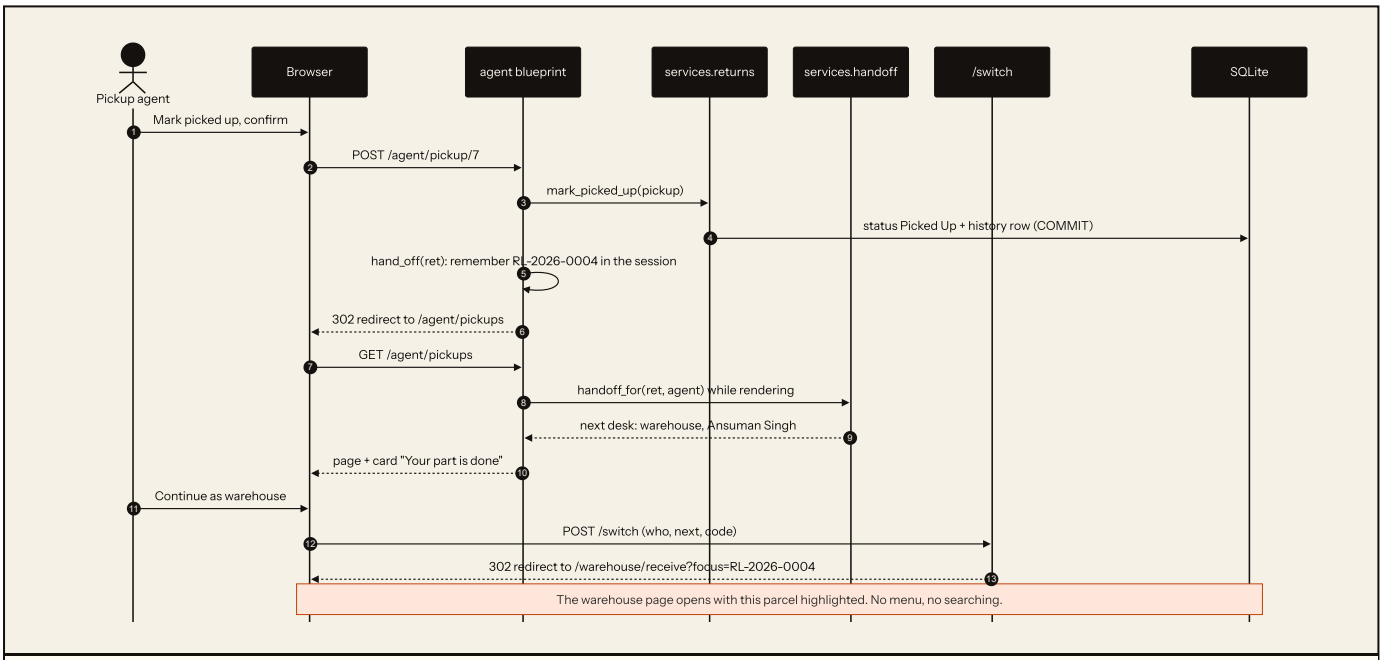


FIGURE 4.8b Sequence: handing a return to the next desk

05 UI design: wireframes & prototypes

5.1 Design language

"Mailroom meets control tower": kraft paper, ink and one signal orange. Return codes and labels are set in a monospace font like shipping labels; grades are rubber stamps; every status badge carries text, so colour is never the only signal.

Paper	Kraft	Ink	Signal orange	Grade A
#F6F1E7	#E9DFCB	#14110F	#FF5B1F	#2F7D4F

Grade B	Grade C
#D99A1E	#B3392B

Bricolage Grotesque headings · **Instrument Sans** body · DM Mono codes & labels

5.2 Wireframes

Low-fidelity sketches of the main screens, drawn before the visual design: layout, content and the one main action on each screen.



FIGURE 5.1 Customer · My orders

Each delivered item is a shipping slip with its return window and one button.



FIGURE 5.2 Customer · Request return

The item on the left; reason tiles, comment and an optional photo on the right.

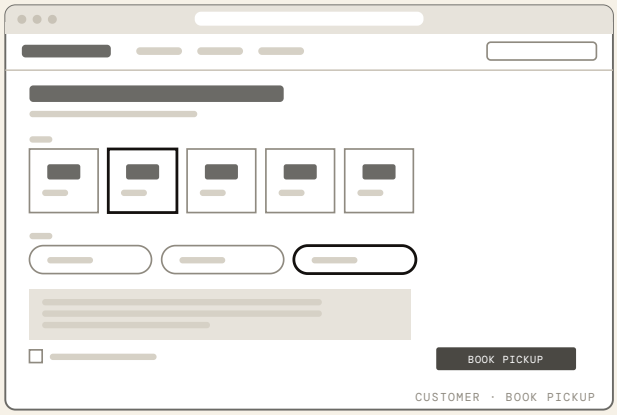


FIGURE 5.3 Customer · Book pickup

Five day tiles, three slots, the pickup address to confirm.

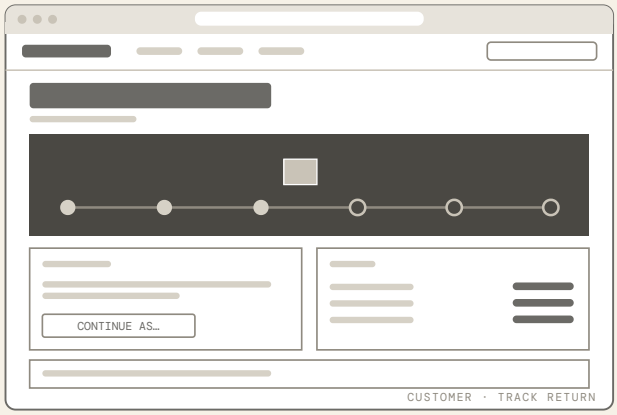


FIGURE 5.4 Customer · Track return

Six-step progress, what happens next, refund breakdown, history.

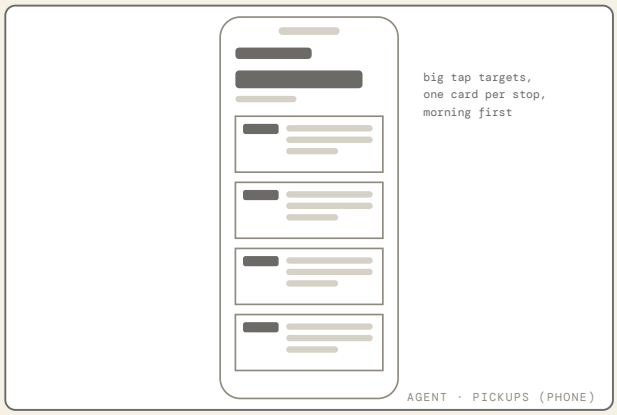


FIGURE 5.5 Agent · Pickups (phone)

Phone-first list of stops, morning first, large tap targets.

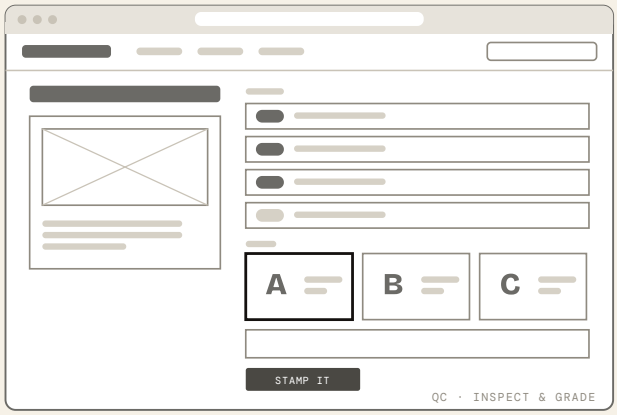


FIGURE 5.6 QC · Inspect & grade

Checklist toggles suggest a grade; the inspector stamps A, B or C.

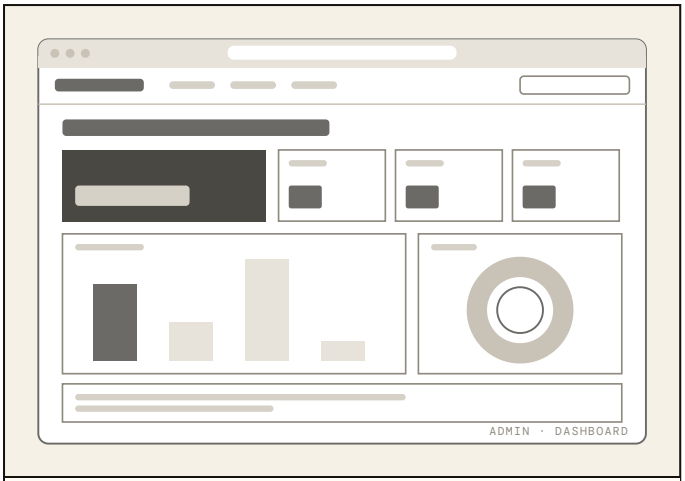


FIGURE 5.7 Admin · Dashboard

Key numbers on top, reasons and grades charts, flagged customers.

5.3 Prototypes

High-fidelity screens of the working prototype, the app as built. Each follows its wireframe above.

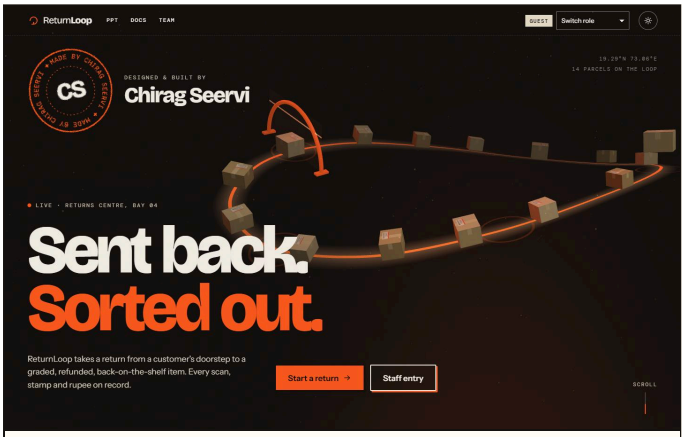


FIGURE 5.8 Landing page

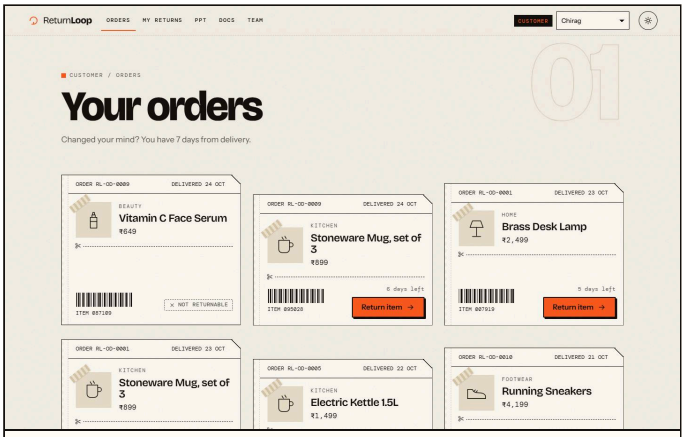


FIGURE 5.9 Customer · My orders

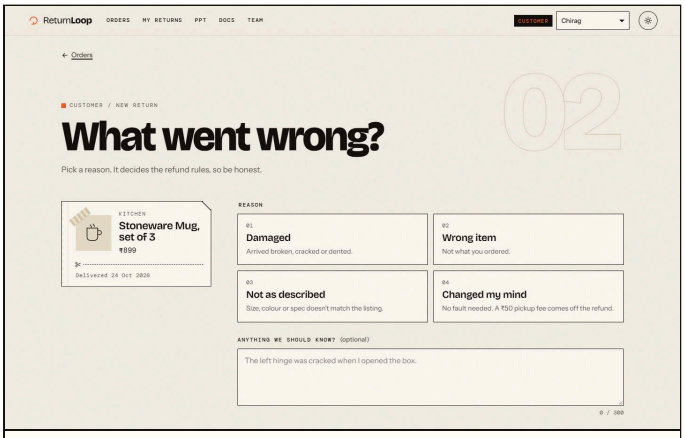


FIGURE 5.10 Customer · Request return

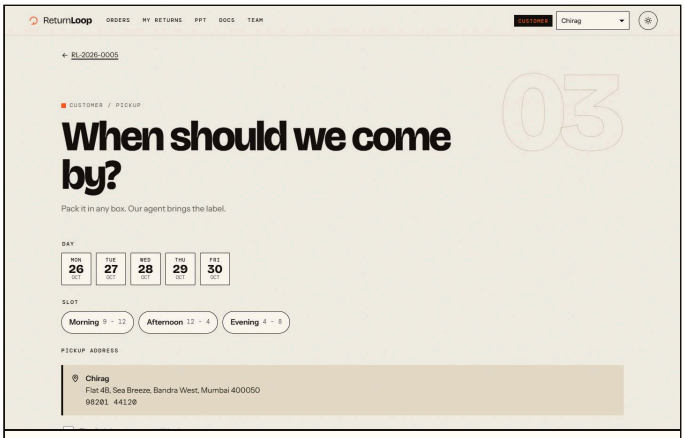


FIGURE 5.11 Customer · Book pickup

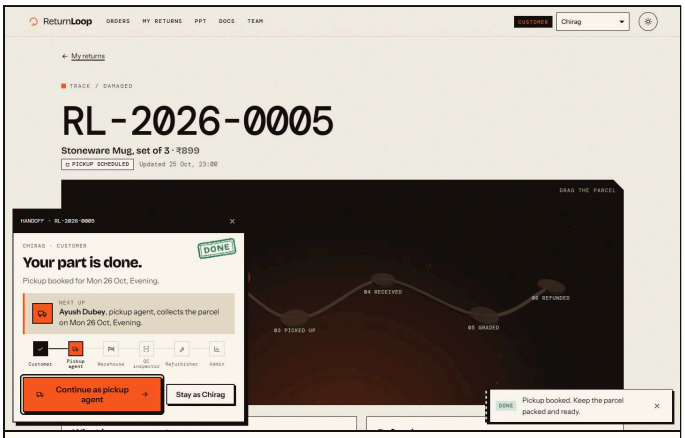


FIGURE 5.12

Customer · Track return, with the handoff card

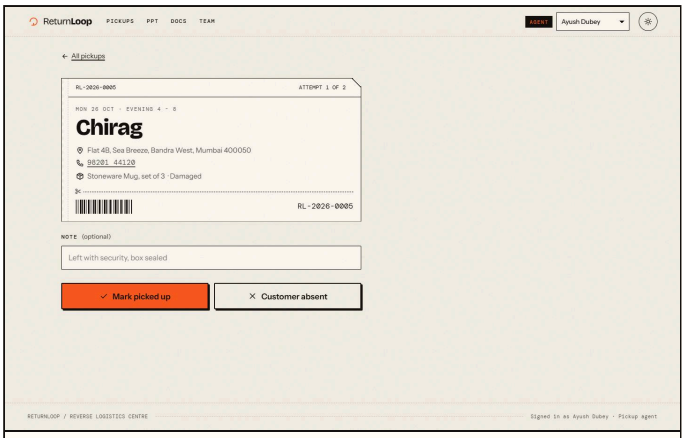


FIGURE 5.13 Agent · Pickup slip

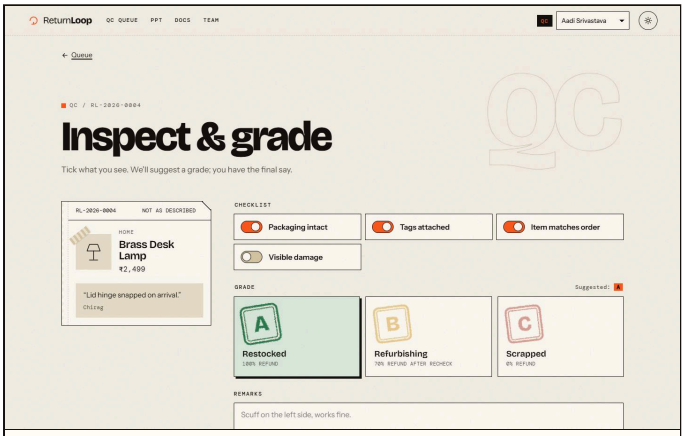


FIGURE 5.14 QC · Inspect & grade

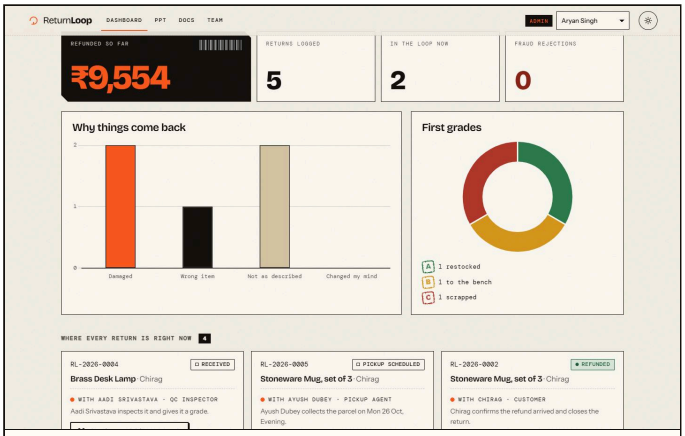


FIGURE 5.15 Admin · Dashboard

RETURNLOOP · SOFTWARE REQUIREMENTS & DESIGN · v1.0

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